

# NPTTEL

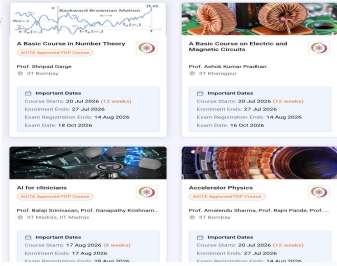
NATIONAL PROGRAMME ON  
TECHNOLOGY ENHANCED LEARNING



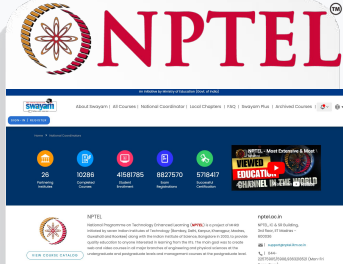
## Faculty / TA Orientation Workshop

July 2026

# NPTEL ONLINE CERTIFICATION (NOC), SINCE 2014



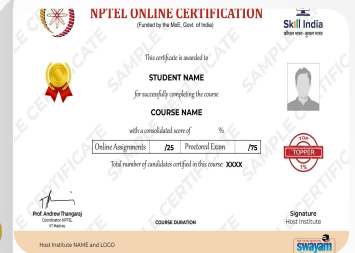
**10,286**  
Courses offered  
on Swayam



**4.16 Crore**  
Enrollments  
(No fee)



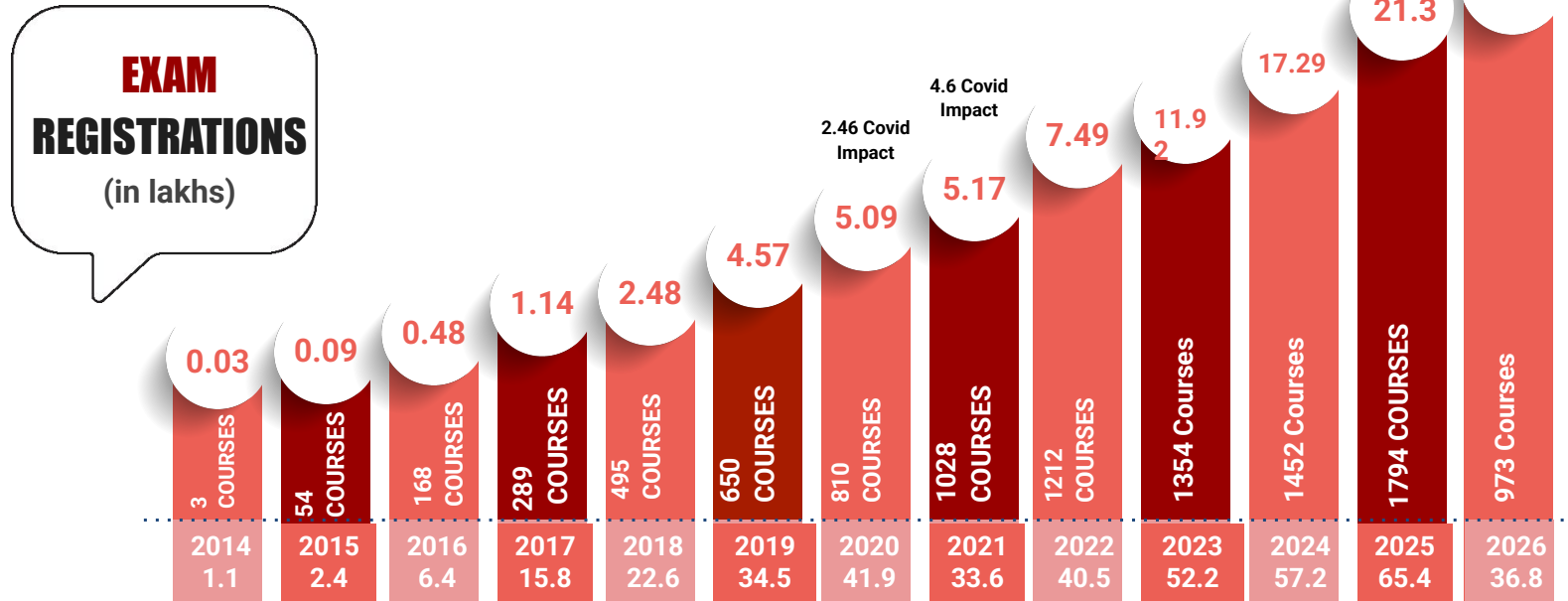
**88.27 Lakhs**  
Exam registrations  
Proctored Exam (Optional)



**57.2 Lakhs**  
Certified  
e-verifiable Certificates

# FROM MAR 2014 – JUN 2026

Jan 2026 Sem: 36.8+ lakhs enrollments,  
12.6+ lakhs exam registrations



**ENROLLED (in lakhs)**

Jul-Dec 2026  
1054 Courses

Enrollments  
6.40 lakhs+  
(Ongoing)

Exam registration  
(Ongoing)

# A Summary of NPTEL Activities

- Largest Contributor on SWAYAM
  - 1054 courses proposed for July 2026 semester
- Running of **SWAYAM portal**
  - Zero downtime since NPTEL took over SWAYAM portal operations
  - Cost at ~Re. 1 per enrolled student per month
- Manage the SWAYAM **local chapter** colleges
  - 8400+ colleges/universities are local chapters
  - Local chapter has roles on portal for monitoring course participation
  - Local chapter SPOC workshops held every semester across IITs
- Regional language **translation**
  - 80063+ translated hours for 3000+ courses

# Factors Contributing to Success of NPTEL

- **Faculty-centric support** for content creation, portal, course delivery
- Active, effective **monitoring of course delivery**
  - Weekly assignments in all courses
  - Dashboard with assignment submissions and grades
  - Forum and assignment data monitoring
- **Examinations:** Verify outcomes, establish value for certification
  - Quality control is a critical part of NPTEL's success
  - Steady growth in exam registration
- **Academic rigour:** Important consideration for credit transfer
- **Community**
  - Active involvement of local chapters (SPoC and Mentors)
  - Acknowledgement of well-performing local chapters

# INTERNSHIPS FOR TOPPERS



**Course offering faculty have to consent to take interns during the summer or winter**



**NPTEL will reach out to the toppers of the course asking them to apply - resume, SOP**



**Faculty chooses from the pool**



**NPTEL sends confirmation**



**Internship - in person at the institute/online**



**Stipend and certificate from NPTEL on successful completion of internship**



## Internships-Summary

| Status    | Winter<br>2022 | Summer<br>2023 | Winter<br>2023 | Summer<br>2024 | Winter<br>2024 | Summer<br>2025 | Winter<br>2025               | Summer<br>2026   |
|-----------|----------------|----------------|----------------|----------------|----------------|----------------|------------------------------|------------------|
| Applied   | 50             | 235            | 103            | 1275           | 300            | 1817           | 3477                         | 4363             |
| Joined    | 44             | 195            | 76             | 204            | 140            | 406            | 1682<br>(Offered)            | 452<br>(Offered) |
| Completed | 44             | 144            | 68             | 92             | 85             | 319            | 820<br>(Completed<br>so far) | Ongoing          |

# 1. POST BAC FELLOWSHIPS - OPPORTUNITY TO WORK WITH TOP FACULTY AT IITS

## Fellowship Details:

- **Duration:** 1 year
- **Eligibility:** NPTEL stars
- **Stipend:** Rs 25,000+HRA per month for stars with UG/PG degrees
- **Contingency** of Rs 1 lakh for 1 year
- **Location:** In-person at the chosen institute

## Project Selection Process:

- Collect project opportunities and academic requirements from the Faculties
- Shared the compiled list of projects with NPTEL stars.

**Student Selection:** Faculty will go through the resumes and videos submitted by the learners and may conduct an interview to select the candidates.



## POST BAC FELLOWSHIPS - PROJECT DETAILS (INSTITUTE-WISE)

| Institutes       | No. of Projects |            |            |
|------------------|-----------------|------------|------------|
|                  | 2024            | 2025       | 2026       |
| IISc Bangalore   | 3               | 7          | 30         |
| IIT Bombay       | 8               | 7          | 34         |
| IIT Delhi        | 4               | 1          | 12         |
| IIT Guwahati     | 15              | 17         | 69         |
| IIT Kanpur       | -               | 7          | 23         |
| IIT Kharagpur    | 33              | 22         | 104        |
| IIT Madras+Other | 12              | 36         | 193        |
| IIT Roorkee      | 28              | 18         | 86         |
| <b>Total</b>     | <b>103</b>      | <b>115</b> | <b>551</b> |

**Total students  
Applied for 2026:  
1041**

# PMRF Research Scholars – Value Addition as NPTEL TAs

- PMRF Scholar's role is in addition to regular NPTEL TAs
  - **Online Live interaction** with course students
  - **Problem solving** and doubt-clearing for weekly content
- History
  - Piloted in Jan-April 2022, with 80 PMRF fellows – Well-received by students
  - Expanded to 370 (July 2022), 466 (Jan 2023), 371(July 2023), 418(Jan 2024), 395(July 2024), 362(Jan 2025), 245(July 2025), 195(Jan 2026)
  - Overwhelming response to the live sessions and mostly positive learner feedback
- Some norms
  1. Part of their PMRF work: No additional honorarium
  2. Does not replace the existing TA system
  3. Conduct live sessions each week for the course she/he is allotted.
  4. Can also work as regular NPTEL TA (with honorarium)
  5. Faculty can opt to not have any PMRF TA
  6. More than one PMRF TA for large classes

# Copyright Guidelines

## NPTEL contents are distributed under the CC-BY-NC-SA license(Attribution-NonCommercial-ShareAlike)

- You can use the material from another source (image, video, article, book, etc.) under the following terms
  1. **Attribution** - You must give appropriate credit , provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
  2. **NonCommercial** - You may not use the material for commercial purposes.
  3. **ShareAlike** - If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.

# Copyright Guidelines

- **Avoiding plagiarism:** Acknowledge the original source
- **Avoiding copyright violation:** Permission to reuse or reinterpret original content
- **Journal Papers:** Please link the DOI or the original Journal site
- **Video content**
  - Do not embed other videos (YouTube or otherwise)
  - Provide link instead, so that learners can view them independently
  - Video can be used after appropriate copyright permissions from the original source

# Course and its contents

# Types of Assignment Questions: Auto-Graded

1. Multiple choice / Multiple select questions (MCQ / MSQ)
2. Numerical (Fill in the blanks / Objective / Short Answer)
3. Alphanumeric (Fill in the blanks / Objective / Short Answer)
4. Numeric Array
5. Programming assignments

# Types of Assignment Questions: Auto-Graded

1. **Multiple choice / Multiple select questions (MCQ / MSQ)**
  - a. Single correct (MCQ) or multiple correct (MSQ) answers possible
  - b. Avoid using “none of the above” / “all of the above”
  - c. We recommend **five or more choices** (random guess  $\leq 20\%$ ) in MCQ, when possible
2. Numerical (Fill in the blanks / Objective / Short Answer)
3. Alphanumeric (Fill in the blanks / Objective / Short Answer)
4. Numeric Array
5. Programming assignments

# Types of Assignment Questions: Auto-Graded

1. Multiple choice / Multiple select questions (MCQ / MSQ)
2. **Numerical (Fill in the blanks / Objective / Short Answer)**
  - a. Answer: Exact-type or Range-type
  - b. Range is preferred to allow for approximations
  - c. Choose the range based acceptable errors (e.g., Range: 1.71 to 1.73, if  $\pm 0.01$ )
  - d. Ensure sufficient range to accommodate reasonable approximations
3. Alphanumeric (Fill in the blanks / Objective / Short Answer)
4. Numeric Array
5. Programming assignments



# Types of Assignment Questions: Auto-Graded

1. Multiple choice / Multiple select questions (MCQ / MSQ)
2. Numerical (Fill in the blanks / Objective / Short Answer)
3. **Alphanumeric (Fill in the blanks / Objective / Short Answer)**
  - a. Assignments: Exact Match | Contains | Starts with | Comma-separated keywords
  - b. Exams: Only exact match is available
4. **Numeric Array**
  - a. Use this to configure vector / matrix / array type answers
5. **Programming assignments**
  - a. Supports: C, C++, Java, Python, Verilog, Perl, Haskell

# Types of Assignment Questions: Manual Grading

## Submission options

- Document upload
- Essay-type questions

## Grading options

- Manual grading
- Peer-reviewed / Rubric-based grading

## How Manual Grading works

- Add **Course Staff** before making the assignment public
  - Course staff are responsible for evaluating the subjective assignments
  - Evaluation deadline is two weeks after assignment due date
  - Course Staff must enter total marks out of 100, before evaluation deadline
- Scores entered on the portal will be visible to the students

Avoid subjective assignments for the first two weeks of the course

# NPTEL Online Courses: Do's and Don'ts for NPTEL Assignments

## Things to Avoid in Assignments (and what to do instead)

- Most common error: Giving Incorrect Answers as Correct Answers

1. Which of the following are prime numbers  
(a) 2 (b) 3 (c) 10 (d) 15

- Another common error: Not giving correct option in MCQ

2. Which of the following are prime numbers  
(a) 4 (b) 6 (c) 10 (d) 15

- ✓ Please get the questions independently verified by a TA.

For example, if TA-1 has not formulated the questions and/or configured on NPTEL website, then ask TA-1 to solve and verify the answers on NPTEL website.

- Please avoid excessive use of “none/all of the above”

3. Which of the following are prime numbers  
(a) 2 (b) 3 (c) All of the above (d) None

- ✓ Use “All” or “None” only for pedagogical reasons.

- Avoid using “All” and “None” in the same question
- Avoid using it when you don't have a good third/fourth choice

- Numerical questions: Prefer giving a range when possible

4. The value of  $\sin(\pi)$  is

Ans. ~~Numerical~~: 0 (Avoid, as  $\sin(3.1415)$  in calculator is not 0)

Ans. Range: -0.001, 0.001

## Things to Avoid in Assignments (and what to do instead)

- ✓ Good practice: is to sort the solutions in either ascending or descending order

- Common causes of confusion for students: Ambiguous; Using qualifiers like “most”/“some”; or rely on personal experience

5. Which number is not identified as prime by most students?  
(a) 2 (b) 3 (c) 10 (d) 15

Be careful when preparing such question. It may be confusing for two reasons: (i) “some” or “most” makes it ambiguous; and (ii) some students may mark 15 as “not identified as prime” (experience based!)

- Text-based auto-graded questions: Be careful!!

6. Name the capital of India

Ans. New Delhi (Think about extra spaces, capitalization, etc.)

- ✓ Consider using regular expressions for such questions

Regular Expression Answer: delhi or Delhi

Note: any solution containing Delhi is marked right (incl. Purani Delhi :-)

- Using text answers for numerical questions to be avoided

7. The result of vector multiplication:  $v = 2*[1 \ 2 \ 3]$  is:

Ans. Text: [2, 4, 6]

- ✓ Use “Question Group” instead:

7. The value of  $v(1)$  is: \_\_\_\_\_. 2

8. The value of  $v(3)$  is: \_\_\_\_\_. 6

# Rescoring of Assignments

**Summary:** Too many errors in the assignments configured on the online Portal

**Solution:** TAs to carefully check all solutions as configured on the portal

Give enough time for NPTEL office to configure and TAs to verify

Why?

- Errors take time to handle
  - For each error: (i) The assignment is made private; (ii) solution are changed; (iii) re-grading job is started; (iv) announcement is posted
- Errors are expensive
  - Approximately Rs. 25,000 to 50,000 per week
- Errors are a reflection of our quality
  - When students ask queries about their exams, they say that if there were errors in the assignments, how sure are we that exam questions are error-free.

# MATLAB software in your course

- MathWorks, the developer of MATLAB and Simulink, is partnering with NPTEL to provide support for online courses.
- MathWorks will collaborate with NPTEL course teams on their online courses that include programming tasks.
- This support includes video tutorials for basic MATLAB skills, technical support on discussion forums, and a free MATLAB license for the duration of the course.
- This has worked quite well in the last 13 semesters.

Incase, if your course will require access to Matlab software and we can put you in touch with the Mathworks team who will guide you on the setup and access and also help with forum questions related to this.

We have sent out an email already on this, Please revert to us if your course required MATLAB Access.

# Best Practices

# Best Practices: Videos and Interactions

## For every 10 minutes of the video

- Create 2-3 questions that the learner can use for self evaluation
- Will be configured below the video and not graded but just used for self test.

## Office Hours

- One hour per week on a live session (with consent of instructor) to answer questions about the assignment that just got closed or about upcoming assignments
- Faculty available online during these “office hours” for doubt clearing.

## Discussion Forum

- Introduce yourself on the forum
- Check the forum twice everyday and answer the queries raised
- If there are not many queries, ask questions to start discussion.

# Best Practices: Weekly Assignments

## Assignments

- Please have TAs check carefully before assignments go live
- If there are errors / queries after the deadline, please address them quickly
- Numeric Answers: Use a reasonably wide range to accommodate rounding errors
- All assignments must be multiple submission enabled.



# Best Practices in Descriptive Courses (1 of 2)

Descriptive courses do not lend themselves as easily to the large scale and auto-graded format of online courses.

But there are a lot of best practices we have learnt over the last eight years

- Use a greater share of MSQs than MCQs
- Use more choices in MCQs (Caveat: Ensure these are real choices)
- Comprehension questions and Grouped questions are very powerful!
  - Use comprehension passage followed by a few linked questions
  - Identify parts of an image / Match-the-columns
  - Email [nkaisare@nptel.iitm.ac.in](mailto:nkaisare@nptel.iitm.ac.in) for suggestions on your favorite question

New question types can be added, but with a lead time of about 6 months!

# Best Practices in Descriptive Courses (2 of 2)

## Essay-type questions

We normally have a few essay type questions on the final exam paper. Avoid surprises so that students stand a good chance to succeed

- Announce this in advance
- Configure essay-type questions in a few assignments to get them to practice
- Prefer essay-type questions in the latter half of your course
- Create some notes to share as sample answers expected for the essay type questions in the course.

# **Timelines and Requirements for July 2026 Semester**

# Contents go live every Friday

We will release content every Friday, so that students get one more weekend to work on the material

Please send the assignments well in advance

- Remember, there are 1054 courses
- We get some of the content only on Sunday
  - Our team works to configure and release on Monday mornings
  - No time for testing and hence prone to errors

Updating the course:

- Add new content (recent advances/new topics) as relevant as required in the course.
- If course is relevant to GATE, a few questions can be tagged so & shared with learners
- If theory course has a lab component, the experiments can be recorded and put up as part of the course content

# Some additional tips

## **Tips to help students succeed**

- Have question types similar to what would be asked in the exam.
- Learners are unable to cope when all assignments are MCQ but final exam is descriptive or paper-pen exam

## **New features will be added to the portal**

- We are working on enabling questions with parameterized values
- Each learner will get a different numerical value for the same question
- Open for suggestions that can be implemented on the portal

# Timelines and Checks when Setting up your QPs

## **Help us complete the exam-setting exercise on time**

- Please send the 2 sets of question papers as per the deadlines given
- Out of 1054 courses 374 courses are in Repeat mode
- For remaining 680 courses, 1360 papers need to be created
- Team of 10 works on creating the QPs on the TCS authoring platform

## **Addressing some concerns of NPTEL exams**

- Check the papers for correctness, missing data, etc.
- Avoid repeating questions across papers
- Check for repeated MCQ options, image clarity, ambiguity in QP, etc.
- Other concerns: (i) Papers having very few questions; (ii) Questions with more than 3 marks each

**Please revert to requests for clarification and sign-off quickly**

## Did you know about Exodus at 90 minutes?

There is 70% exodus in every exam session at 90 minutes after exam start

Papers with 20 or 30 multiple choice questions are completed very quickly.  
Think “Kaun banega crorepati”

If question paper does not have numerical problems or comprehension problems, increase the questions to 100 so that learners at least have to answer more

Can be repetitions from assignments with minor tweaks

# Publicity

## **1. Posters and booklet**

We have published the course posters and working on the booklets for publicity and will be disseminating to 8400+ colleges we work with.

## **2. Campaigns on FB and Instagram**

We have been advertising for the courses on the NPTEL social media platforms and running paid campaigns too on them.

## **3. Workshops**

We will organize awareness workshops in collaboration with colleges in different states and doing these online.

### **On your part:**

1. You can publicise for the same on any forums you may be a part of - professional, social etc.
2. Request you to also publicise among any industry contacts you may have, your students' network, etc
3. In case of visiting any college for any talks too, would be great if you can probably talk about NPTEL and the availability of these courses
4. If you have contact with print/tv media, you can try for publicity on this front too.



# **Question Papers and NPTEL QP Tool**

# QP Requirements

## QP FORMAT:

**4 Weeks** - Question paper must be for a total of 50/100 marks. (Engage the learner for at least 1.5 hrs)

**8/12 Weeks** - Question paper must be for a total of 100 marks. (Engage the learner for at least 2-2.5 hrs)

### **Questions Split up (A MUST)**

- **Recall** : Questions from the videos of your course
- **Assignments** : Questions similar to your weekly assignments
- **Open** : Questions to test the understanding of the learner and differentiate the students based on how much they have learned. Can be inferential questions or advanced level too.

## TYPE OF EXAM:

- Mention the type of exam (Online/Offline)

### **Online exam:**

Objective type paper - all questions to be answered on the computer - most of the questions can be auto graded.

### **Offline Exam:**

Questions will be displayed in the monitor and candidates need to write their answer in the provided answer sheet. Once we receive the answer papers, we shall send the same to you for manual evaluation.

## NUMBER OF QUESTION PAPERS:

- Required two sets of encrypted final exam question paper (Either word/Pdf).
- The encrypted papers should be mailed ONLY to Ms. Bharathi and the password for the same should be sent to her mobile no (9884686692) via WhatsApp/sms.

# ADVANTAGES OF QP TOOL

- Questions will always be available in the tool for future semesters for rerun courses.
- From the QP bank, previous semester questions can be pulled and added to the current semester.
- All the types of questions are available in the QP tool.
- Both the shifts can be created simultaneously and in parts also

*Thank you!*